# Supplementary Online Appendix to:

**Changing Global Linkages: A New Cold War?**

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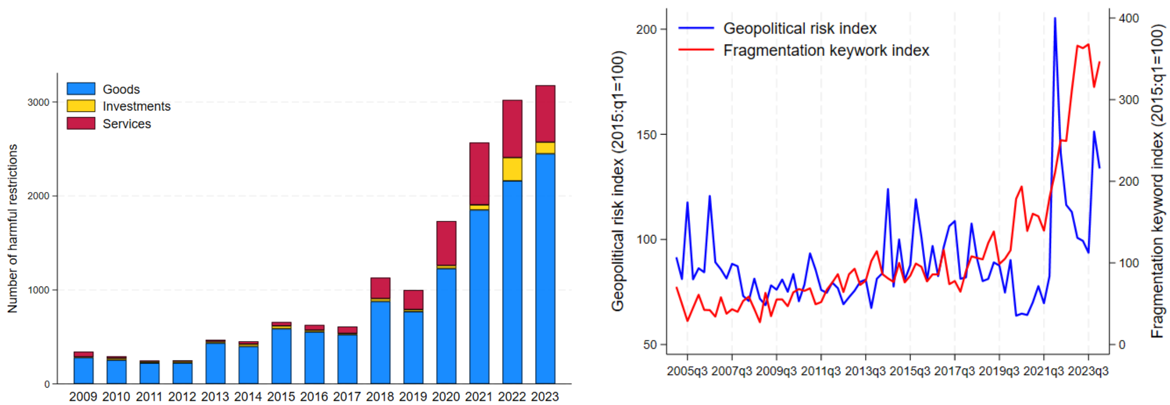
## S1 Additional figures

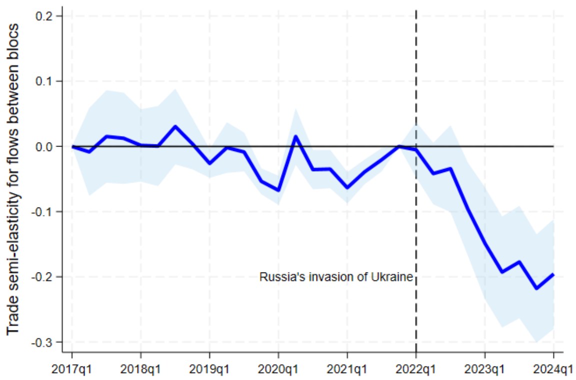
Figure S1.1: Rising Fragmentation Pressures



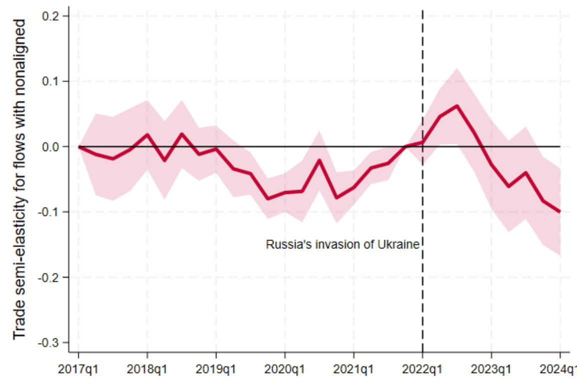

(a) Harmful Restrictions on Trade and Investment (b) Geopolitical risk and Fragmentation indexes

Notes: Panel A plots the number of harmful restrictions on trade and investment per year. Panel B plots the geopolitical risk developed by Caldara and Iacoviello (2022) (Data downloaded from https://www.matteoiacoviello.com/gpr.htm on September 2024) and the fragmentation risk index, which measures the average number of sentences, per thousand earnings calls, that mention at least one of the following keywords: deglobalization, reshoring, onshoring, nearshoring, friend-shoring, localization, regionalization.
Sources: Caldara and Iacoviello (2022); Global Trade Alert; NL Analytics; and authors' calculations.

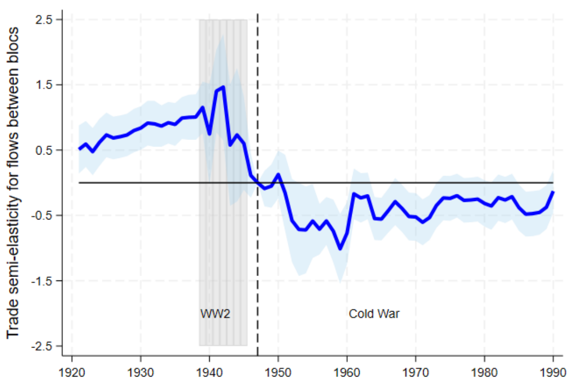
Figure S1.2: Timing of Trade Fragmentation: The Cold War and Now



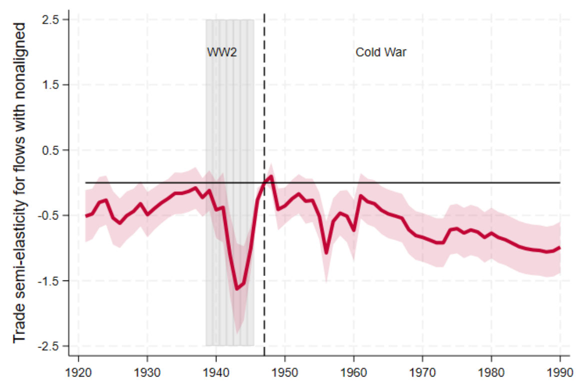

(a) Trade between blocs, now




(b) Trade with nonaligned, now



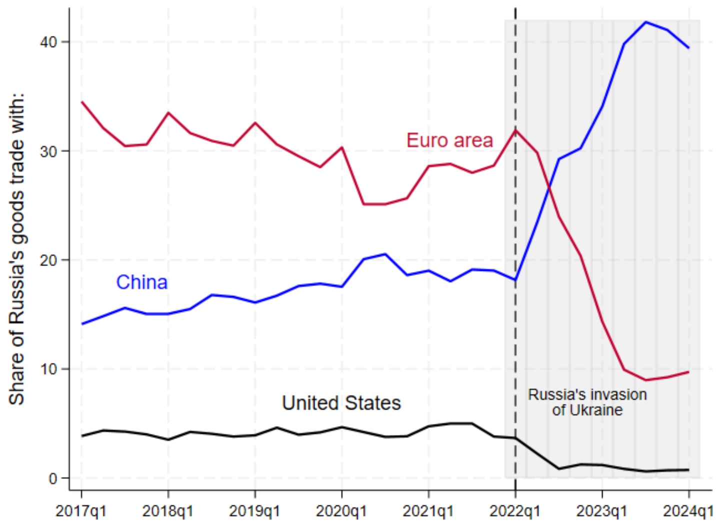

(c) Trade between blocs, Cold War




(d) Trade with nonaligned, Cold War

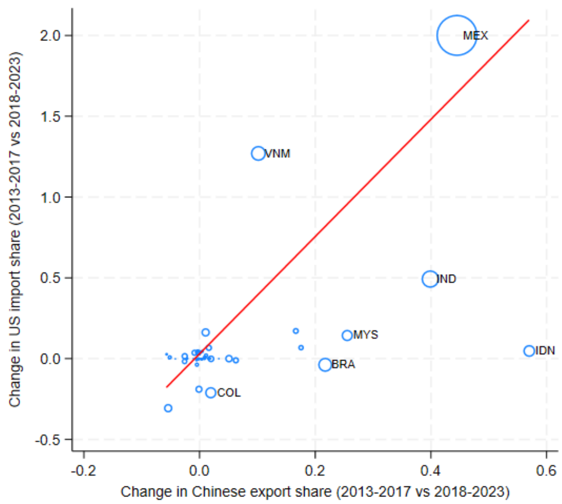
Notes: The charts plot global trade between blocs (panels A and C) and with nonaligned (panels B and D) around the Russian invasion of Ukraine (panels A and B) and the Cold War (dated as starting in 1947). For each episode, the chart plot the semi-elasticity of trade for flows between blocs and with nonaligned, and the associated 90 percent confidence bands, estimated with PPML and a fully saturated gravity model as in equation 1 in the main text. The missing category is trade within blocs. The Cold War results are obstained using yearly data from 1920 to 1990—with 1947 as excluded year—and the bloc definition based on Gokmen (2017). The results for the most recent period are based on quarterly trade data from 2017:Q1 to 2024:Q1 (with 2021:Q4 as excluded quarter), with the wider bloc definition based on the Ideal Point Distance (a measure based on voting pattern in the UNGA computed by Bailey *et al.* (2017)).
Sources: Trade Data Monitor; Historical Bilateral Trade and Gravity dataset (TRADHIST) prepared by Fouquin and Hugot (2016); and authors' calculations.

Figure S1.3: Change in Russia's Trade Flows

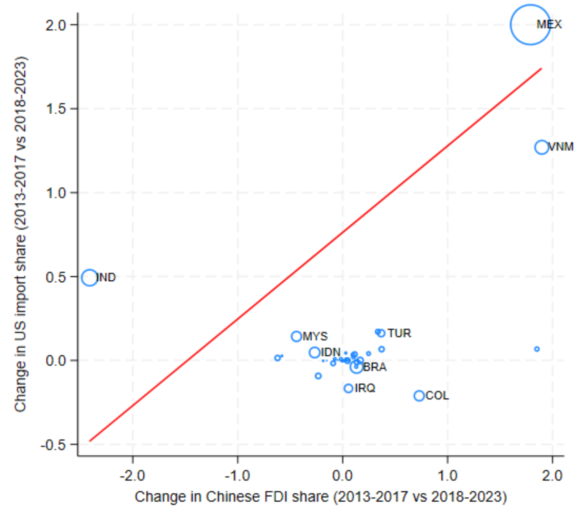



Notes: The chart plots the share of Russia's goods trade with China, U.S. and the Euro area, before and after the Russia's invasion of Ukraine.
Sources: Trade Data Monitor; and authors' calculations.

Figure S1.4: The Emergence of Connector Countries: Excluding Japan



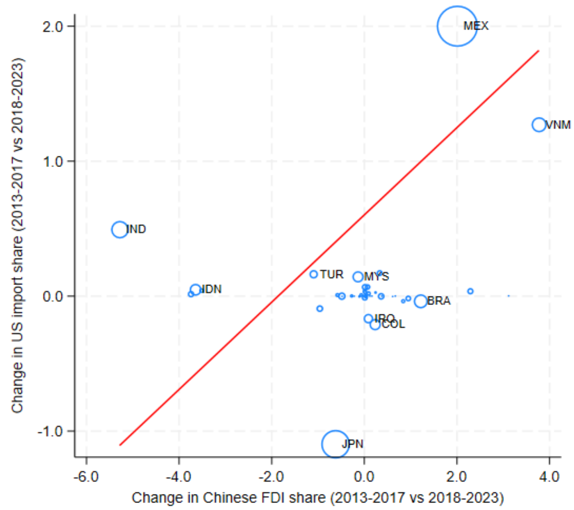

(a) Change in US import shares vs Chinese export shares




(b) Change in US import shares vs FDI from China

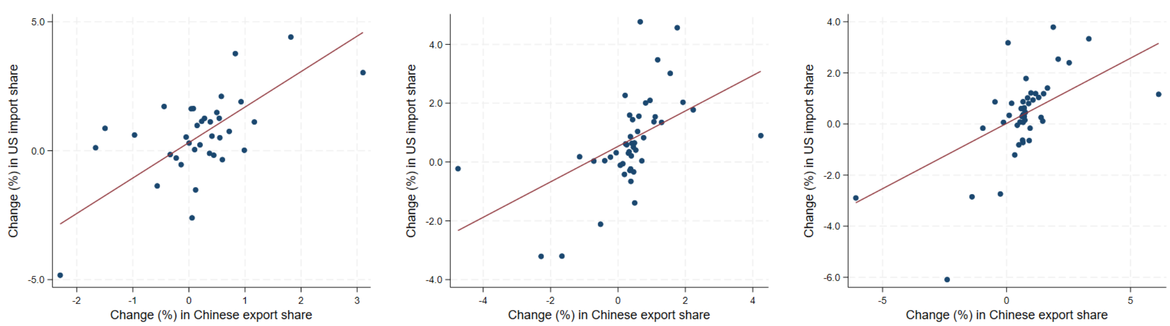
Notes: All panels include only nonaligned countries. Panel A plots the change in US import shares between 2018-2023 and 2013-2017 against the change in Chinese export shares over the same period. A weighted regressions (with the US imports in the pre-period as weights) with robust standard errors gives a slope equal to 3.628 (p-value = 0.001); n = 56. Panel B plots the change in US import shares between the period 2018-23 and the period 2013-17 against the change in Chinese outward FDI over the same period. A weighted regressions (with the US imports in the pre-period as weights) with robust standard errors gives a slope equal to 0.516 (p-value = 0.013); n = 44.
Sources: Trade Data Monitor; fDi Markets; and authors' calculations.

Figure S1.5: The Emergence of Connector Countries: Value Instead of Count FDI Data




Notes: The chart plots the change in US import shares between the period 2018-23 and the period 2013-17 against the change in Chinese outward FDI over the same period, using values (in USD). A weighted regressions (with the US imports in the pre-period as weights) with robust standard errors gives a slope equal to 0.323 (p-value = 0.092); n = 45. Sources: Trade Data Monitor; and authors' calculations.

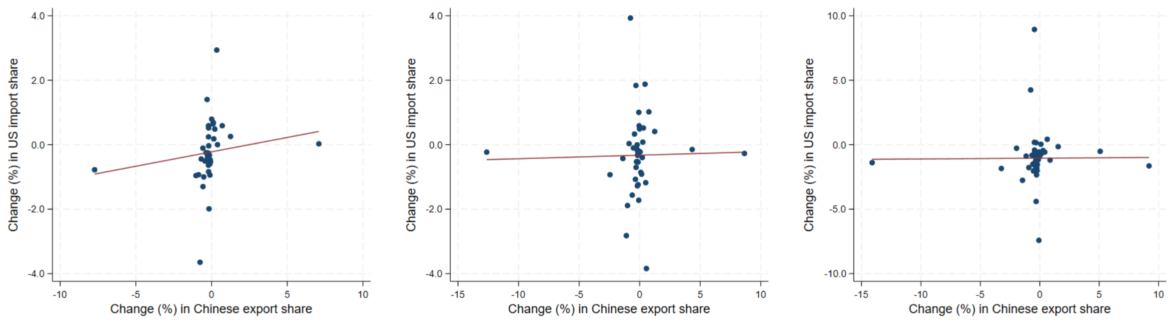
Figure S1.6: The Emergence of Connector Countries: Product-level Analysis




(a) HS2 product level (b) HS4 product level (c) HS6 product level

Notes: All panels include only nonaligned countries and products (defined at the HS 6-digit level) targeted by the tariffs imposed by the US administration on Chinese imports in 2018 and 2019. All panels plot the change in US import shares between 2018-2023 and 2013-2017 against the change in Chinese export shares over the same period. Panel A defines products at the 2-digit HS level, while panels B and C at the 4-digit and 6-digit HS levels, respectively. The charts are binned scatterplots that absorb product-level fixed effects and correspond to the results reported in Table S2.4, panel A, columns 1, 3 and 5.
Sources: UN Comtrade; and authors' calculations.

Figure S1.7: The Emergence of Connector Countries: Placebo Test




(a) HS2 product level    (b) HS4 product level    (c) HS6 product level

Notes: All panels include only nonaligned countries and products (defined at the HS 6-digit level) targeted by the tariffs imposed by the US administration on Chinese imports in 2018 and 2019. All panels plot the change in US import shares between 2013-2015 and 2010-2012 against the change in Chinese export shares over the same period. Panel A defines products at the 2-digit HS level, while panels B and C at the 4-digit and 6-digit HS levels, respectively. The charts are binned scatterplots that absorb product-level fixed effects and correspond to the results reported in Table S2.4, panel B, columns 1, 3 and 5.
Sources: UN Comtrade; and authors' calculations.

## S2 Additional tables

Table S2.1: Reallocation across Import and FDI Sources: Evolution over Time Periods

Notes: The table reports the coefficients from regressing the Lilien Index computed on the value of imports (panel A) and FDI (panel B) on indicators for different time periods, with 2003-2007 being the excluded period. Column (1) includes all countries in the sample, column (2) only advanced economies, while column (3) includes only emerging markets and developing economies. In column (4) the regression is based on the sample of countries hypothetically aligned with the U.S./EUR, while the remaining countries are in column (5). Standard errors in parenthesis are clustered at the country level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

| | (1)<br>All | (2)<br>AEs | (3)<br>EMDEs | (4)<br>US bloc | (5)<br>Others |
|---|---|---|---|---|---|
| | | | **Panel A. Trade flows** | | |
| 2008-2012 | 0.3388** | 0.4303*** | 0.2727 | 0.3975*** | 0.2856 |
| | (0.135) | (0.123) | (0.189) | (0.120) | (0.200) |
| 2013-2021 | 0.0736 | 0.4728*** | -0.1270 | 0.2748 | -0.0622 |
| | (0.119) | (0.165) | (0.155) | (0.164) | (0.163) |
| 2022-2023 | 0.9899*** | 1.4318*** | 0.7507*** | 1.5973*** | 0.6116** |
| | (0.209) | (0.243) | (0.285) | (0.245) | (0.294) |
| Observations | 2,639 | 785 | 1,854 | 893 | 1,746 |
| $R^2$ | 0.491 | 0.447 | 0.455 | 0.403 | 0.463 |
| | | | **Panel B. FDI flows** | | |
| 2008-2012 | 0.1136 | -1.0572 | 0.9119 | -0.5016 | 0.6639 |
| | (0.654) | (0.930) | (0.870) | (0.870) | (0.953) |
| 2013-2021 | 0.4424 | -0.6077 | 1.1596 | -0.3118 | 1.0977 |
| | (0.672) | (1.039) | (0.865) | (0.919) | (0.964) |
| 2022-2023 | 2.8543*** | 5.6496*** | 1.1920 | 4.8882*** | 1.2284 |
| | (0.978) | (1.787) | (1.041) | (1.459) | (1.258) |
| Observations | 1,504 | 577 | 927 | 665 | 839 |
| $R^2$ | 0.287 | 0.285 | 0.298 | 0.274 | 0.302 |
| Country FE | Y | Y | Y | Y | Y |

Table S2.2: Trade, Investment and Portfolio Flows: Excluding the U.S. and China

Notes: The table reports the results of a gravity model using Poisson pseudo-maximum likelihood (PPML, columns 1-4, 7-8) and ordinary least squares (OLS, columns 5-6), where the dependent variable is: the bilateral trade in US dollars (columns 1-2 and 7-8); the number of announced FDI projects (columns 3-4); and the change in the share of portfolio assets (columns 5-6). Data are quarterly in columns 1-4, semiannual in columns 5-6, and annual in columns 7-8. The sample spans 2017:q1-2024:q1 (columns 1-2), 2008:q1-2024:q2 (columns 3-4), 2015:s1-2023-s2 (columns 5-6) and 1920-1990 (excluding World War II, 1939-1945, columns 7-8). The FDI analysis excludes country pairs with less than 5 investment projects over the sample period. Panel A excludes the U.S. from the sample, while panel B excludes China. The Post War variable identifies the period following Russia's invasion of Ukraine and is equal to 1 from 2022:q1 onwards. The Cold War dummy is equal to 1 for the years 1947-1990. The Between Bloc variable equals 1 if the source and destination country do not belong to the same geopolitical bloc, and 0 otherwise. The Nonaligned variable equals 1 if at least one country in the pair is nonaligned. In the current period, blocs are centered around the U.S. and China, with a group of countries remaining nonaligned. The bloc definition uses the Ideal Point Distance (a measure based on UNGA voting patterns computed by Bailey *et al.* (2017)) to assign countries into blocs. Standard errors in parenthesis are clustered at the source-destination pair level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
|---|---|---|---|---|---|---|---|---|
| | Trade around the Russian invasion of Ukraine | | FDI around the Russian invasion of Ukraine | | Portfolio holdings around the Russian invasion of Ukraine | | Trade during the Cold War | |
| | Panel A. Wider bloc definition, excluding the U.S. | | | | | | | |
| Between Bloc × Post War | -0.0904** | -0.0864** | -0.3690*** | -0.2039*** | -0.0121 | -0.0324* | -0.3957*** | -1.0171*** |
| | (0.038) | (0.043) | (0.066) | (0.071) | (0.016) | (0.017) | (0.142) | (0.111) |
| Nonaligned × Post War | 0.0710** | 0.0329 | -0.0166 | -0.1986*** | 0.0034 | -0.0269 | -0.2546** | -0.5554** |
| | (0.031) | (0.042) | (0.038) | (0.071) | (0.015) | (0.022) | (0.116) | (0.223) |
| Observations | 254,688 | 254,572 | 157,872 | 139,263 | 230,639 | 230,622 | 749,554 | 672,430 |
| | Panel B. Wider bloc definition, excluding China | | | | | | | |
| Between Bloc × Post War | -0.1340** | -0.1300** | -0.3051*** | 0.0598 | -0.0422 | -0.0570* | -0.7904*** | -1.1139*** |
| | (0.059) | (0.054) | (0.103) | (0.079) | (0.026) | (0.030) | (0.198) | (0.119) |
| Nonaligned × Post War | 0.0453 | 0.0145 | 0.0070 | -0.0472 | -0.0212 | -0.0356 | -0.2596** | -0.4705** |
| | (0.028) | (0.041) | (0.046) | (0.074) | (0.021) | (0.032) | (0.110) | (0.236) |
| Observations | 254,660 | 254,589 | 160,512 | 142,297 | 230,656 | 230,622 | 754,667 | 676,457 |
| Country-pair FE | Y | Y | Y | Y | Y | Y | Y | Y |
| Time FE | Y | - | Y | - | Y | - | Y | - |
| Source × Time FE | N | Y | N | Y | N | Y | N | Y |
| Destination × Time FE | N | Y | N | Y | N | Y | N | Y |

Table S2.3: FDI Flows Between Blocs: Value Instead of Count Data

Notes: The table reports the results of a gravity model using Poisson pseudo-maximum likelihood where the dependent variable is the value (in USD) of announced FDI projects. Data are quarterly and the sample spans 2008:q1-2024:q2 and exclude country pairs with less than 5 investment projects over the sample period.The Post War variable identifies the period following Russia's invasion of Ukraine and is equal to 1 from 2022:q1 onwards. The Between Bloc variable equals 1 if the source and destination country do not belong to the same geopolitical bloc, and 0 otherwise. The Nonaligned variable equals 1 if at least one country in the pair is nonaligned. Blocs are centered around the U.S. and China, with a group of countries remaining nonaligned. The bloc definition uses the Ideal Point Distance (a measure based on UNGA voting patterns computed by Bailey *et al.* (2017)) to assign countries into blocs. Standard errors in parenthesis are clustered at the source-destination pair level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

| | (1) | (2) |
|---|---|---|
| | FDI around the Russian invasion of Ukraine | |
| Between Bloc × Post War | -0.7613*** | -0.3411** |
| | (0.166) | (0.162) |
| Nonaligned × Post War | -0.2104* | 0.0883 |
| | (0.112) | (0.236) |
| Observations | 170,412 | 152,088 |
| Country-pair FE | Y | Y |
| Time FE | Y | - |
| Source × Time FE | N | Y |
| Destination × Time FE | N | Y |

Table S2.4: The Emergence of Connector Countries: Product-level Analysis

Notes: The table reports the results of the OLS estimation of the following regression:

$$\Delta\ USimport\ share_{ip} = \beta_1 \Delta\ China\ export\ share_{ip} + \beta_2 Growth_i + \phi_p + \epsilon_{ip},$$

where the dependent variable is the percentage change in US import shares computed between the period 2018-2023 and the period 2013-2017 (panel A) or between 2015-2013 versus 2012-2010 (panel B) of country $i$ and product $p$. The main explanatory variable is the percentage change in Chinese export shares computed over the same periods. Even columns also control for the ex-ante average real GDP growth in country $i$ (computed over the period 2013-2017 in panel A and 2010-2012 in Panel B). The model also include product fixed effects ($\phi_p$). Regressions are weighted using the US imports in the pre-period as weights. The sample includes only nonaligned countries and products (defined at the HS 6-digit level) targeted by the tariffs imposed by the US administration on Chinese imports in 2018 and 2019. The sample also excludes country-product pair that account for less than 0.5% of total US imports of that products in the post-period. Columns 1-2 define products at the 2-digit HS level, while columns 3-4 and 5-7 at the 4-digit and 6-digit HS levels, respectively. Column 7 reports results separately for products (defined at the 6-digit HS level) hit or not by the 2018-2019 US tariffs. Standard errors in parenthesis are clustered at the country-product pair level. Annual bilateral product-level trade data are from UN Comtrade. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

| Dependent variable: | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
|---|---|---|---|---|---|---|---|
| Change (%) in US import share | HS2 level | | HS4 level | | HS6 level | | |
| **Panel A. Main analysis: Change between 2018-2023 versus 2013-2017** | | | | | | | |
| Change (%) in Chinese export share | 1.3758*** | 1.2361*** | 0.6028*** | 0.5048*** | 0.5108*** | 0.4329*** | |
| | (0.404) | (0.425) | (0.206) | (0.195) | (0.161) | (0.159) | |
| Change (%) in Chinese export share × Tariff=0 | | | | | | | -0.1038 |
| | | | | | | | (0.148) |
| Change (%) in Chinese export share × Tariff=1 | | | | | | | 0.4538*** |
| | | | | | | | (0.166) |
| Growth | | 0.3909*** | | 0.5680*** | | 0.6874*** | 0.6796*** |
| | | (0.056) | | (0.137) | | (0.126) | (0.125) |
| Observations | 1,308 | 1,308 | 7,685 | 7,685 | 22,021 | 22,021 | 22,021 |
| $R^2$ | 0.578 | 0.646 | 0.611 | 0.640 | 0.656 | 0.674 | 0.675 |
| **Panel B. Placebo: Change between 2013-2015 versus 2010-2012** | | | | | | | |
| Change (%) in Chinese export share | 0.0893 | 0.0770 | 0.0109 | -0.0044 | 0.0059 | -0.0098 | |
| | (0.070) | (0.065) | (0.021) | (0.021) | (0.021) | (0.021) | |
| Change (%) in Chinese export share × Tariff=0 | | | | | | | -0.0061 |
| | | | | | | | (0.013) |
| Change (%) in Chinese export share × Tariff=1 | | | | | | | -0.0201 |
| | | | | | | | (0.063) |
| Growth | | 0.1583** | | 0.3614*** | | 0.5033*** | 0.5030*** |
| | | (0.062) | | (0.082) | | (0.130) | (0.130) |
| Observations | 1,162 | 1,161 | 6,438 | 6,436 | 17,016 | 17,015 | 17,015 |
| R-squared | 0.233 | 0.263 | 0.515 | 0.541 | 0.674 | 0.685 | 0.685 |
| Product FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Weights (US import in the pre-period) | Yes | Yes | Yes | Yes | Yes | Yes | Yes |

## S3 Reliability of the fDi Markets data

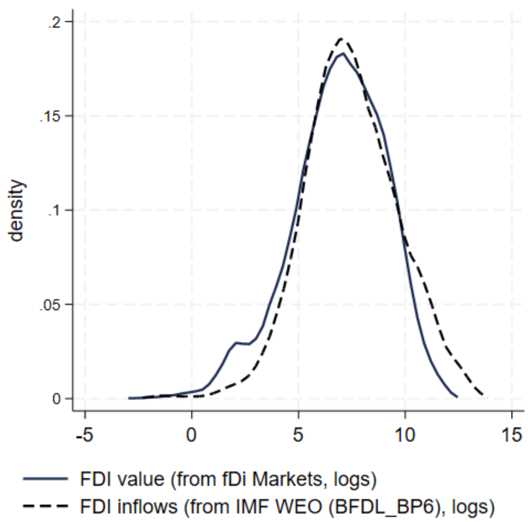
FDI data from fDi Markets track announcements of new and expansions of existing projects that create jobs and capital investment. fDi Markets does not track mergers and acquisitions and other international equity investments, projects that do not create new jobs, or companies, which establish a foreign subsidiary without a physical presence. The data are collected primarily from publicly available sources (e.g., media, industry organizations, investment promotion agency newswires) and report investment-level information for over 330,000 FDI instances between January 2003 and June 2024 between 186 countries. As the value of capital investment is often estimated rather than directly reported, our analysis is mostly based on the number of FDI projects. However, we establish the robustness of our main results to the use of FDI values.

Toews and Vézina (2022) and Aiyar *et al.* (2024) discuss the reliability of the fDi Markets data by comparing the value at the destination country-year level computed from fDi Markets with gross FDI inflows from the IMF's balance of payment statistics. They show: (i) a strong positive correlation between the number of fDi investment projects and value of FDI from balance of payment statistics; and (ii) a strong positive correlation between the count and (estimated) value of investment projects in fDi Markets.

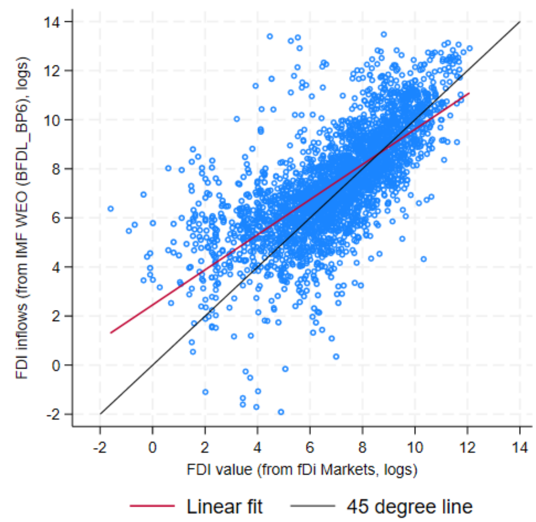
To establish the reliability of the data sourced from fDi Markets, we apply the same analysis to our sample and report a set of statistics that corroborates the quality of the FDI count data. We aggregate the volumes at the destination country-year level and contrast them with gross FDI inflows as published in the World Economic Outlook (WEO) over the period 2003-2023. Figure S3.1 shows that the two distributions present a large degree of overlap (Panel a) and the two sets of data are highly correlated (in Panel b the slope of the linear fit is equal to 0.72 and the $R^2$ is 0.50). In addition, the count and volume of bilateral investment are highly correlated (in Panel c the slope of the linear fit is equal to 0.96 and the $R^2$ is 0.66), supporting the choice of using the count data in our baseline analysis.

Finally, to mitigate the possibility that our findings are affected by phantom FDI, we exclude FDI from and to offshore financial centers and tax havens, as defined by Damgaard *et al.* (2024).

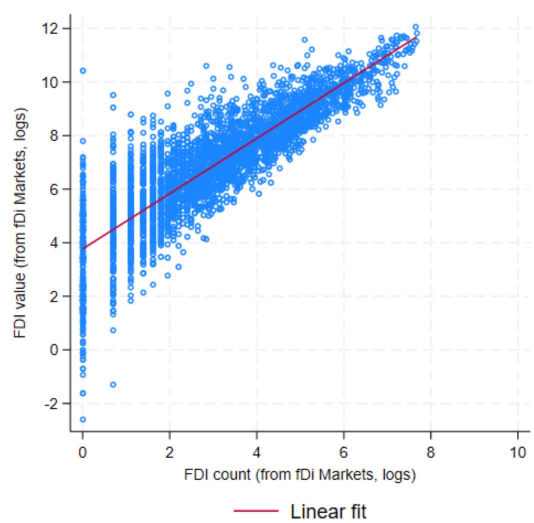
Figure S3.1: FDI data: Micro vs Macro and Count vs Value




(a) Investment-level and aggregate FDI flows, densities




(b) Investment-level and aggregate FDI flows




(c) FDI count (number of individual investments) vs FDI value

Notes: Panels a) and b) plot FDI inflows at the country-year level from the IMF World Economic Outlook database and from the fDi Market database between 2003 and 2023 in USD. Both variables are expressed in logarithms. In panel b) the $R^2$ of the estimated linear fit is 0.50 and the slope is 0.72 (s.e. = 0.01). Panel c) plots FDI inflows at the country-year level only from fDi Market between 2003 and 2023, comparing flows in value and in number of investments; the $R^2$ of the estimated linear fit is 0.66 and the slope is 0.96 (s.e. = 0.00). Both variables are expressed in logarithms.

## S4 Additional references

CALDARA, D. and IACOVIELLO, M. (2022). Measuring geopolitical risk. *American Economic Review*, **112** (4), 1194–1225.

DAMGAARD, J., ELKJAER, T. and JOHANNESEN, N. (2024). What is real and what is not in the global FDI network? *Journal of Interational Money and Finance*, **140**, 102971.

TOEWS, G. and VÉZINA, P.-L. (2022). Resource Discoveries, FDI Bonanzas, and Local Multipliers: Evidence from Mozambique. *The Review of Economics and Statistics*, **104** (5), 1046–1058.